Numerical Linear Algebra for Computational Science and Information Engineering
CITHN2006

Retake Final Exam

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Problem 1 (2 pts)

Consider the matrix

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}.$$

a. Is A normal, or not? Explain. (1 pt)

> Since A is skew-symmetric, i.e., $A^T = -A$, we have:
>
> $$AA^T = A(-A) = (-A)A = A^T A$$
>
> so that A is normal.

b. What is the conditioning number of solving for any eigenvalue of A? (1 pt)

> Since A is normal, each of its pairs of unit right- and left-eigenvectors, denoted by u and v, is aligned. Then, solving for any simple eigenvalue λ of A is characterized by a conditioning number $\kappa(A, \lambda) = 1/|u^H v| = 1/|u^H u| = 1$. That is, solving for a simple eigenvalue λ of A is well-conditioned, irrespective of λ.

Problem 2 (4 pts)

Let $P \in \mathbb{R}^{n\times n}$ be any orthogonal projector, i.e., $P^T = P$. We show that P has a unit 2-norm, i.e., $\|P\|_2 = 1$, as follows:

a. Show that $\|x\|_2^2 = \|Px\|_2^2 + \|(I_n - P)x\|_2^2$ and $\|Px\|_2 \leq \|x\|_2 \,\forall\, x \in \mathbb{R}^n$. (2 pt)

Since P is an orthogonal projector, for all $x \in \mathbb{R}^n$, we have:

$$\begin{aligned}
\|x\|_2^2 &= \|Px + (I_n - P)x\|_2^2 \\
&= \|Px\|_2^2 + \|(I_n - P)x\|_2^2 + x^T P^T (I_n - P)x + x^T (I_n - P)^T Px \\
&= \|Px\|_2^2 + \|(I_n - P)x\|_2^2 + x^T P^T x - xP^T Px + x^T Px - x^T P^T Px \\
&= \|Px\|_2^2 + \|(I_n - P)x\|_2^2 + x^T Px - x^T P^2 x + x^T Px - x^T P^2 x \\
&= \|Px\|_2^2 + \|(I_n - P)x\|_2^2 + x^T Px - x^T Px + x^T Px - x^T Px \\
&= \|Px\|_2^2 + \|(I_n - P)x\|_2^2
\end{aligned}$$

so that $\|x\|_2^2 \geq \|Px\|_2^2$ and thus $\|Px\|_2 \leq \|x\|_2$.

b. Using the definition of the matrix 2-norm, show that $\|P\|_2 \geq 1$. (1 pt)

As a non-trivial projector, P is such that there exists $z \neq 0$ such that $Pz = z$, i.e., $\exists\, z \in \text{range}(P)$. Then, given the definition of the matrix 2-norm:

$$\|P\|_2 = \sup_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\|Px\|_2}{\|x\|_2},$$

we have:

$$\|P\|_2 \geq \frac{\|Pz\|_2}{\|z\|_2} \geq \frac{\|z\|_2}{\|z\|_2} \geq 1.$$

c. Using the result of a. with the definition of the matrix 2-norm, show that $\|P\|_2 \leq 1$. (1 pt)

From a., we have $\|Px\|_2 / \|x\|_2 \leq 1 \,\forall\, x \in \mathbb{R} \setminus \{0\}$ so that, using the definition of the matrix 2-norm, we get

$$\|P\|_2 = \sup_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\|Px\|_2}{\|x\|_2} \leq 1.$$